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Pharmacy Expiry & Re-order Dispatch Engine

Lab 1: Requirements Engineering & UML Use-Case Modelling

1. Problem Context & Overview

Hospital pharmacies require an automated stock management engine that tracks medicine batch expiry dates, supports First-Expired-First-Out (FEFO) dispensing, and automatically initiates purchase orders when stock falls below a configured safety threshold.

Primary stakeholders/actors: Pharmacy Clerk and Inventory Supplier.

2. Requirements Table

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall enforce First-Expired-First-Out (FEFO) picking by prioritizing the valid medicine batch with the nearest expiry date during dispensing.	High	Pass: When multiple valid batches exist, the batch with the nearest expiry date is selected first. Fail: A newer-expiring batch is selected while an older valid batch exists.	Prevents medicine wastage and ensures proper expiry-date management.
FR-002	Functional	The system shall allow the Pharmacy Clerk to add and update medicine batch details, including medicine name, batch number, quantity, expiry date, and safety-stock threshold.	High	Pass: A batch is successfully stored or updated when all required fields contain valid values. Fail: The system accepts incomplete or invalid batch information.	Accurate batch data is necessary for expiry tracking and inventory management.
FR-003	Functional	The system shall identify and flag expired medicine batches and prevent them from being selected for dispensing.	High	Pass: A batch whose expiry date has passed is marked as expired and cannot be selected for dispensing. Fail: An expired batch remains available for dispensing.	Prevents dispensing expired medicines.
FR-004	Functional	The system shall monitor the available stock quantity of	High	Pass: When stock quantity becomes lower than the	Ensures timely replenishment and reduces the risk of

		each medicine and detect when the quantity falls below its configured safety-stock threshold.		threshold, the system marks the medicine as requiring re-order. Fail: A below-threshold medicine is not detected.	stock shortages.
FR-005	Functional	The system shall automatically generate and dispatch a purchase order request to the Inventory Supplier when the stock quantity falls below the configured safety-stock threshold.	High	Pass: A purchase order containing the required medicine and quantity is generated and dispatched after a below-threshold condition is detected. Fail: No purchase order is generated or dispatched.	Automates the re-order process and helps maintain medicine availability.
NFR-001	Non-Functional (Performance & Security)	The system shall automatically generate and dispatch supplier purchase order requests when item stock dips below the safety threshold.	High	Pass: Benchmarking tests confirm the target latency and security standards under simulated peak load. Fail: The system does not meet the defined performance or security benchmark.	Ensures reliable and secure automated supplier communication.
NFR-002	Non-Functional (Reliability)	The system shall preserve medicine batch and purchase-order data during normal operation and system recovery.	High	Pass: After a simulated restart or recovery, previously saved batch and purchase-order records remain available and unchanged. Fail: Stored records are lost or corrupted.	Pharmacy inventory and purchase-order records must remain reliable.

3. Actors and Responsibilities

Actor	Responsibilities
Pharmacy Clerk	Adds and updates medicine batches, generates FEFO dispensing lists, views expiry information, and monitors stock levels.
Inventory Supplier	Receives dispatched purchase orders and processes the requested medicine supply.
System Administrator	Configures safety-stock thresholds and manages inventory-related system settings.

4. Identified Use Cases

Use Case ID	Use Case	Purpose
UC-01	Manage Medicine Batch	Add or update medicine batch details.
UC-02	Generate FEFO Dispensing List	Prioritize valid medicine batches with the nearest expiry date.
UC-03	Detect Expired Medicine	Identify expired batches and prevent dispensing.
UC-04	Monitor Stock Level	Check available stock against the configured safety threshold.
UC-05	Generate Purchase Order	Create a purchase order when stock falls below the threshold.
UC-06	Dispatch Purchase Order	Send the generated purchase order to the Inventory Supplier.
UC-07	Configure Safety Threshold	Set or update the minimum safe quantity for medicines.

5. UML Use-Case Diagram Specification

System Boundary: Pharmacy Expiry & Re-order Dispatch Engine

Actor Associations:

- Pharmacy Clerk → Manage Medicine Batch
- Pharmacy Clerk → Generate FEFO Dispensing List
- Pharmacy Clerk → Detect Expired Medicine
- Pharmacy Clerk → Monitor Stock Level
- System Administrator → Manage Medicine Batch
- System Administrator → Configure Safety Threshold
- Inventory Supplier → Dispatch Purchase Order

«include» Relationship: Generate Purchase Order «include» Dispatch Purchase Order. This represents mandatory behavior: the generated purchase order must be dispatched to the supplier.

«extend» Relationship: Detect Expired Medicine «extend» Generate FEFO Dispensing List. This represents conditional behavior when expired batches are encountered during FEFO processing.

6. Use-Case Flow Specification

Use Case ID: UC-02

Use Case Name: Generate FEFO Dispensing List

Primary Actor: Pharmacy Clerk

Preconditions:

- The Pharmacy Clerk is authorized to access the system.
- Medicine batch information has already been recorded in the system.
- At least one medicine batch is available in inventory.

Postconditions:

- A dispensing list is generated based on FEFO order.
- The valid batch with the nearest expiry date is prioritized.
- Expired batches are excluded from the dispensing list.

Main Success Scenario:

1. The Pharmacy Clerk selects the “Generate FEFO Dispensing List” option.
2. The system retrieves all available batches for the requested medicine.
3. The system checks the expiry date of each batch.
4. The system identifies and excludes expired batches.
5. The system compares the expiry dates of all remaining valid batches.
6. The system sorts the valid batches according to the nearest expiry date.
7. The system selects the batch with the nearest valid expiry date as the highest-priority batch.
8. The system generates the FEFO dispensing list.
9. The Pharmacy Clerk views the recommended batch for dispensing.
10. The Pharmacy Clerk confirms the medicine dispensing.
11. The system updates the remaining stock quantity of the selected batch.
12. The system checks whether the remaining stock has fallen below the configured safety-stock threshold.
13. If the stock is below the threshold, the system initiates the purchase-order generation process.
14. The use case ends successfully.

Alternate Flow: 4a. No Valid Batch Available

- The system detects that all available batches have passed their expiry dates.
- The system marks the batches as expired.
- The system prevents the Pharmacy Clerk from selecting any expired batch for dispensing.
- The system displays the message: “No valid medicine batch is available for dispensing.”
- The Pharmacy Clerk may check other available inventory or initiate the re-order process.
- The use case ends.